1 comparison 1

2 comparison 2

	invariant model	non-invariant model
equivariant circuit $U(\boldsymbol{\theta})$	**Equiv** $f_{\text{eq}}(x) = \langle\psi(x)\|U^\dagger O\, U\|\psi(x)\rangle$ 2 1	**Equiv-NoSym** $f_{\text{ns}}(x) = \langle\psi(x)\|U^\dagger O_{\text{ns}}\, U\|\psi(x)\rangle$ 2
non-equivariant circuit $V(\boldsymbol{\theta})$	**NonEquiv-Twirled** $f_{\text{tw}}(x) = \frac{1}{8}\sum_g f_{\text{neq}}(R_{\mathcal{X}}(g)x)$ 1	**NonEquiv** $f_{\text{neq}}(x) = \langle\psi(x)\|V^\dagger O\, V\|\psi(x)\rangle$ 1

$$V = C_{\text{c}}\, U C_{\text{c}}^\dagger, \quad O = \tfrac{1}{2}(X_0 + X_4), \quad O_{\text{ns}} = (X_0 + Z_0 + Z_4)/(1+\sqrt{2})$$